
1. load b
2. phase estimation
3. invert
4. uncompute
5. measure
ancilla $|0\rangle$
clock $|0\rangle^{\otimes n}$
memory $|0\rangle^{\otimes m}$
n
m
prepare $|b\rangle$
QPE
$R_y(\theta_k)$
QPE†
keep the run if 1
$|0\rangle^{\otimes n}$
i with probability $\propto |x_i|^2$